MIL-G-50676A (AR)
28 February 1982
SUPERSEDING
MIL-G-50676 (MU)
15 April 1971

MILITARY SPECIFICATION

GATE I-F: 10548649

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, quality assurance provisions, and packaging for an I-F gate assembly, used in Radar Receiver R-1475/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

FSC: 1285

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-46357	-	Computer, Gun Direction; M18 Manufacturing Processes and inspection of
MIL-G-50683	-	Gate, I-f Board Assembly - Upper: 10548656
MIL-G-50684	-	Gate, I-f Board Assembly - Lower: 10548654

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-810	-	Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

F10548649	I-f Gate

PACKAGING DATA SHEET

10548649	Packaging of I-f Gate

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Fabrication. The i-f gate shall be manufactured in accordance with Drawing F10548649 and drawings pertaining thereto. Subassemblies shall have met the requirements of the following specifications:

Subassembly	Specification
I-f Gate Board - Upper: C10548656	MIL-G-50683
I-f Gate Board - Lower: D10548654	MIL-G-50684

3.2 General specifications. Where applicable, the i-f gate shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357.

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. The i-f gate shall be mounted on the vibration table in normal service position. The i-f gate shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for 2 hours along each of three mutually perpendicular axes within the frequency range of 10 to 55 hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency at each frequency range shall not exceed 1 Hz per second. Vibration tests shall be conducted under ambient room conditions.

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3.3.2 Shock. The i-f gate shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 ±1 milliseconds (ms). The G value of each impulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5.5 ms.

3.3.3 Altitude.

3.3.3.1 Operating. The i-f gate shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. The i-f gate shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The i-f gate shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +155°F.

3.3.4.2 Nonoperating (storage). The i-f gate shall not suffer permanent damage when subjected to not less than 4 hours at each temperature extreme of -65°F and +160°F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (nonoperating). The i-f gate shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity of 95 ±5 percent for a period of not less than 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the i-f gate.

3.5 Performance.

3.5.1 Input conditions. For the requirements of 3.5.2, the test setup as specified in Figure 1 with item 7 disconnected from J2 and the following conditions a through c shall apply:

a. 15.0 ±0.1 volts direct current (Vdc) between J5-1(+) and J5-3(-).

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b. 15.0 +0.1 Vdc between J5-5(-) and J5-3(+).
c. Positive pulse of 20 kilohertz (kHz) pulse repetition frequency (PRF), 2.0 microsecond (us) wide and +10 volts (V) peak-to-peak into 50 ohms shall be applied to J1.
d. 60 megahertz (MHz) continuous wave (cw) signal of 650 millivolts (MV) peak-to-peak shall be applied to J2. For the requirements of 3.5.3 and 3.5.4, the test setup as specified in Figure 1 with item 7 connected and the above conditions a through d shall apply.

3.5.2 Gate offset. The dc offset, of the output signal at J3, during the gate on time shall be adjustable by R8 to the same dc level as the dc level during the gate off time.

3.5.3 Gate output. The output signal at J3 shall be attenuated during the 2.0 us pulse period with the following conditions:

a. RF amplitude - not less than 400 MV peak-to-peak.
b. Gate time (off) - not less than 2.02 us, or not more than 2.422 us at the 50 percent points.
c. Fall time (r-f attenuation) - not more than 0.1 us.
d. Rise time (r-f return) - not more than 0.1 us.
e. Amplitude of r-f during gate off - not more than 4.0 millivolts (mV) peak-to-peak.

3.5.4 Modulator trigger output. The pulse measured at J4, when terminated into 50 ohms, shall be the same pulse as specified in paragraph 3.5.1 c, except it may be attenuated not more than 10 percent.

3.6 First Article Inspection. This specification makes provisions for First Article Inspection. Requirements for the submission of First Article Samples by the contractor shall be as specified in the contract.

3.7 Workmanship. All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface blisters, excess flash, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing the base material. The assembly operations shall be performed in a thorough, workmanlike manner consistent with the best practices of the industry. All required markings shall be neat and sharply defined.

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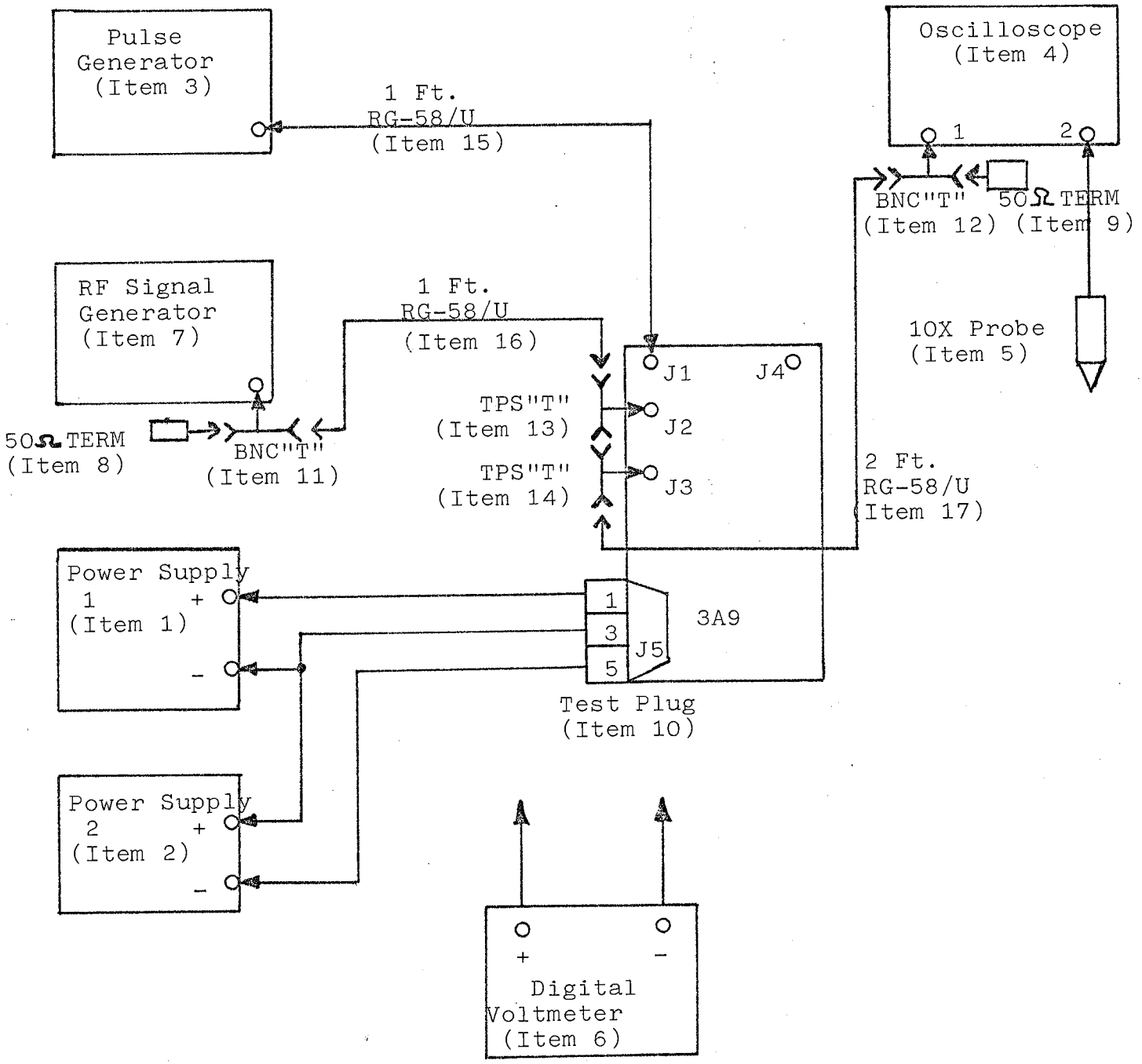


FIGURE 1. I-F Gate test setup

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisons of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) submission. The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three i-f gates manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provision (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristics) basis in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 and 4.4 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.3.2.3 and 4.4 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests. The requirements and tests in Table I shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature (60°F and 90°F).

TABLE I. Classification of defects
(100 percent inspection)

Characteristic	Requirement	Test procedures
Critical: None		
Major:		
101. Electrical	3.4	4.6.2
102. Performance	3.5	4.6.3
Minor: None		

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4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. One i-f gate shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table II. Three i-f gates shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table III.

TABLE II. Special sampling

Control test	Requirement	Test procedures
301. Fabrication	3.1	Applicable drawings-Visual
302. General specifications	3.2	MIL-F-13926 & MIL-C-46357-Visual

TABLE III. Special sampling environmental

Characteristic	Requirement	Test procedure
303. Vibration	3.3.1	4.6.1.1
304. Shock	3.3.2	4.6.1.2
305. Altitude	3.3.3	4.6.1.3
306. Temperature	3.3.4	4.6.1.4
307. Humidity (nonoperating)	3.3.5	4.6.1.5

4.4.1 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Test equipment. In conjunction with 4.5.2.2, the standard test equipmnt listed in Table IV shall be utilized in performing the applicable test of 4.6.

TABLE IV. Test equipment list

Item number	Description
1. Power supply	14 to 16 Vdc regulated, capacity 200 mA, ac ripple less than 10 mV peak-to-peak.
2. Power supply	Same as item 1.
3. Pulse generator	Pulse amplitude greater than +10 Vdc into 50 ohms. 20.0 ±0.2 kHz PRF. Rise and fall times less than 20 ns. Trigger pulse which precedes the above pulse from 1 to 10 us, variable.

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TABLE IV. Test equipment list. continued

Item number	Description
4. Oscilloscope	Bandwidth of 80 MHz with rise time less than 20 us. Sensitivity of 50 mV/cm to 10 V/cm. Time base from 0.1 us/cm to 10 us/cm. Provided with delayed sweep having external synchronization.
5. Oscilloscope probe	Provide an impedance of 10 megohms with less than 10 pF loading. Rise time response of less than 10 ns.
6. Digital voltmeter	0 to 20 Vdc, accuracy ±0.05% of reading.
7. Rf signal generator	60 MHz ±5%. Output amplitude 2.5 Vrms. (variable ±20%) into 50 ohms.
8. Rf termination	50 ±1 ohms. Accurate to 100 MHz.
9. Rf termination	Same as item 8.
10. Test Plug	Mating Plug for connector receptacle 10549074-1.
11. Connector	UG-274/U BNC "T".
12. Connector	Same as item 11.
13. Connector	TPS "T" #G4768-000-000 Automatic Metal Products Corp. 315-323 Berry St., Brooklyn NY 11211 or equivalent.
14. Connector	Same as item 13.
15. Cable	1 ft RG-58/U male BNC to Automatic Metal Products Corp. #G4701-000-880 or equivalent.
16. Cable	Same as item 15.
17. Cable	2 ft RG-58/U male BNC to Automatic Products Corp. #G4701-000-880 or equivalent.

Note: A dual power supply meeting the requirements listed under Item 1 description may be substituted for Items 1 and 2.

4.6 Test methods and procedures.

4.6.1 Environmental. Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

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4.6.1.1 Vibration. The test shall be performed as specified in Method 514, Procedure IX, Part I of MIL-STD-810 except that failure criteria do not apply. After vibration, the i-f gate shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.1.

4.6.1.2 Shock. This test shall be performed as specified in Procedure I, Method 516 of MIL-STD-810 except that shock pulses shall be 11.0 ms and 15 G. After the shock impulses, the i-f gate shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.2.

4.6.1.3 Altitude (nonoperating and operating). This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 feet (5.5 in. Hg). Operation of Step 3 shall be the performance tests of 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and nonoperating). This test shall be performed as specified in Procedure I, Method 501 of MIL-STD-810. The temperature of Step 2 shall be +160°F. The temperature of Step 3 shall be maintained for not less than 4 hours. The temperature of Step 4 shall be +155°F. In Step 5 the temperature shall be maintained for 4 hours and the i-f gate shall be tested in accordance with 4.6.2 and 4.6.3 during this time.

4.6.1.4.2 Low (operating and nonoperating). This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -65°F maintained for not less than 4 hours. The temperature of Step 4 shall be -40°F. In Steps 5 and 7, the i-f gate shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity (nonoperating. This test shall be performed in accordance with Procedure II, Method 507 of MIL-STD-810. The i-f gate shall be tested in accordance with 4.6.3.

4.6.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance.

4.6.3.1 Input conditions. The selected i-f gate assemblies shall be tested as follows: Using test setup shown in Figure 1 and

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input signals as specified in 3.5.1 a through d, check for presence of output signals as specified in 3.5.2 and 3.5.3.

4.6.3.2 Gate offset. Check the dc offset of the output signal as specified in 3.5.2.

4.6.3.3 Gate output. Check for presence of gate output signal as specified in 3.5.2.

4.6.3.4 Modulator trigger output. Check for presence of modulator trigger output signal as specified in 3.5.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548649. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The i-f gate assembly is used in the Radar Receiver R-1475/VPS-2, part of Radar Set AN/VPS-2. The i-f gate assembly attenuates the i-f transmitted burst leakage into the receiving system. If not attenuated the leakage would saturate the receiving system, obscuring close range targets.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable packaging data sheet number (see 5.1).
c. Selection of applicable level of preservation packaging and packing.
d. Level of protection required.
e. Applicable stock number.
f. Requirement for first article submission.

6.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring of the equipment. Partial submission of inspecton equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAF-I, Dover, NJ 07801.

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When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of transmittal:

a. The contract number.
b. The contract item (name, model number, etc.).
c. The designs remaining to be submitted and the expected date of transmittal.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Changes to previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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